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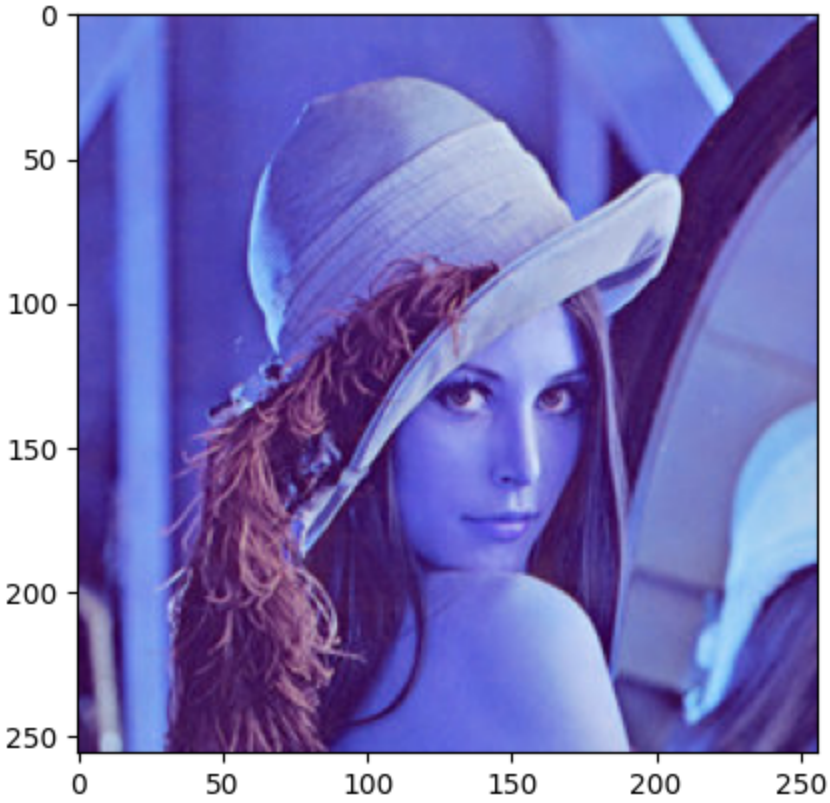
FECHA=26 de Mayo

In [1]:
```
import matplotlib.pyplot as plt
import cv2
import numpy as np

img= cv2.imread('images/Lena_RGB.png')
print('Image dimensions: ', np.shape(img))

plt.imshow(img, cmap='gray')# TRANSFORMACION A GRISES
plt.show()
```

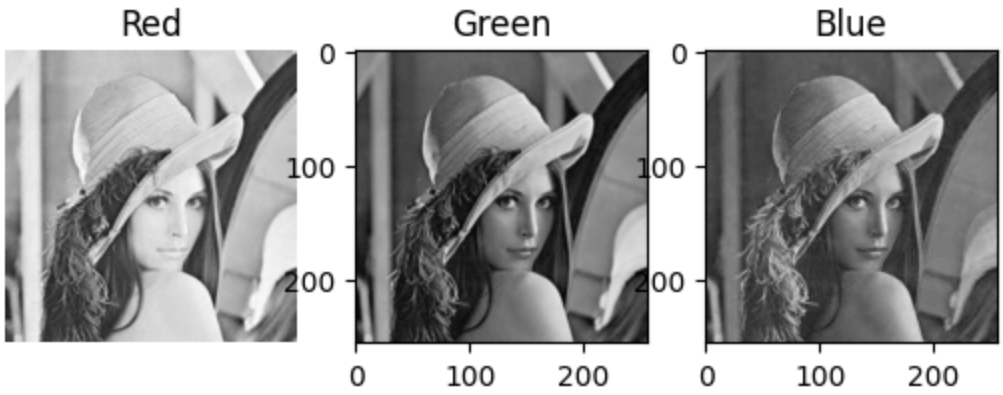
```
Image dimensions:  (256, 256, 3)
```



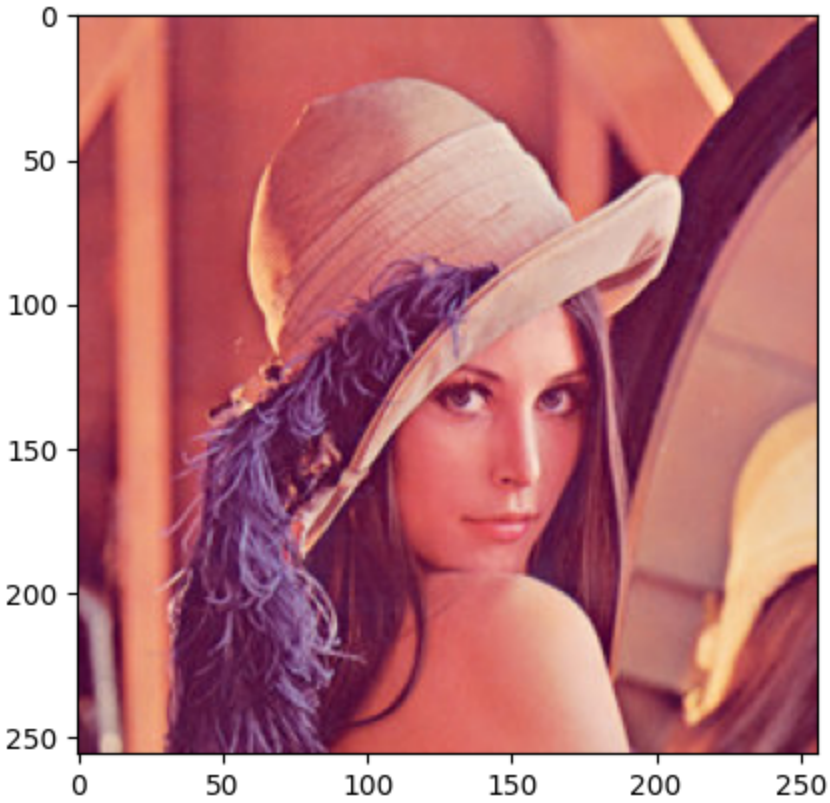
In [2]:
```
# Extraer por separado la imagen de grises de cada canal
R = img[:,:,2]#filas columnas y color
G = img[:,:,1]
B = img[:,:,0]
print (R)
print("color verde")
print(G)
```

```
[[225 225 225 ... 230 229 215]
 [226 225 225 ... 249 239 207]
 [225 225 226 ... 231 213 177]
 ...
 [ 91  95  98 ... 139 157 166]
 [ 90  93  96 ... 156 171 177]
 [ 88  90  93 ... 168 180 185]]
color verde
[[135 135 135 ... 144 142 127]
 [136 135 136 ... 157 147 114]
 [136 136 134 ... 136 119  85]
 ...
 [ 25  30  31 ...  52  59  61]
 [ 24  28  29 ...  63  66  66]
 [ 22  25  26 ...  71  73  70]]
```

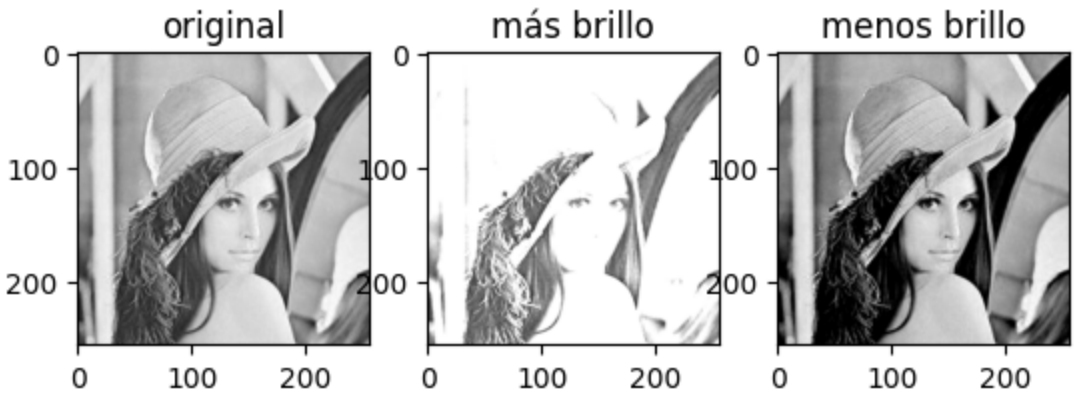
In [3]:
```python
# Visualizar los canales en un subplot
fig, ax = plt.subplots(1,3)
ax[0].imshow(R, cmap='gray'), ax[0].set_title('Red'), ax[0].axis('off')
ax[1].imshow(G, cmap='gray'), ax[1].set_title('Green')
ax[2].imshow(B, cmap='gray'), ax[2].set_title('Blue')
plt.show()
```



In [4]:
```python
RGB_img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB) #CVT COLOR TRASNFROMAR  DE BGR A RGB
plt.imshow(RGB_img, cmap='gray')
plt.show()
```

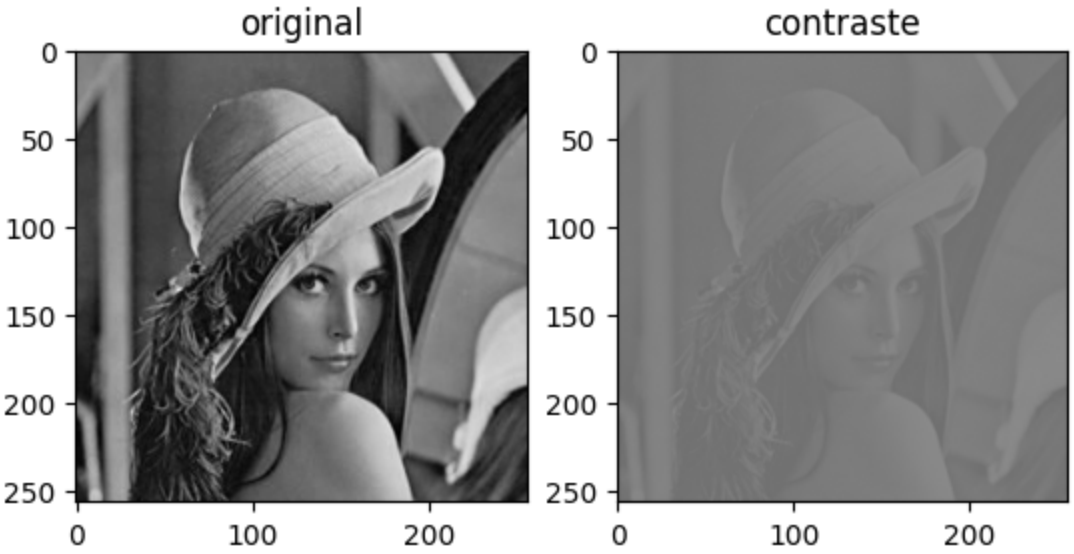


In [5]:
```
# CAMBIO DE BRILLO
img = cv2.imread('images/Lena_RGB.png')
img = img[:,:,2] # red color
mas_brillo = 100
menos_brillo =-100
mas_brillo_img = cv2.add(img, mas_brillo) # Importante el "cv2.add" en vez de
menos_brillo_img = cv2.add(img, menos_brillo)
fig, ax = plt.subplots(1,3)
ax[0].imshow(img, cmap='gray'),
ax[0].set_title('original')
ax[1].imshow(mas_brillo_img, cmap='gray'),
ax[1].set_title('más brillo')
ax[2].imshow(menos_brillo_img, cmap='gray'),
ax[2].set_title('menos brillo')
plt.show()
```



0.0.1 TRANSFORMACIONES DE INTENSIDAD

In [6]:
```
# CAMBIO DE CONTRASTE de acuerdo con el programa GIMP
img = cv2.imread('images/Lena_RGB.png')
img = img[:,:,1] # canal verde

contraste =-100
f = 131*(contraste + 127)/(127*(131-contraste))
alpha_c = f
gamma_c = 127*(1-f)

contrast_img = cv2.addWeighted(img, alpha_c, img, 0, gamma_c)

fig, ax = plt.subplots(1,2)
ax[0].imshow(img, cmap='gray', vmin=0, vmax=255), ax[0].set_title('original')
ax[1].imshow(contrast_img, cmap='gray', vmin=0, vmax=255), ax[1]. set_title('contra
plt.show()
```

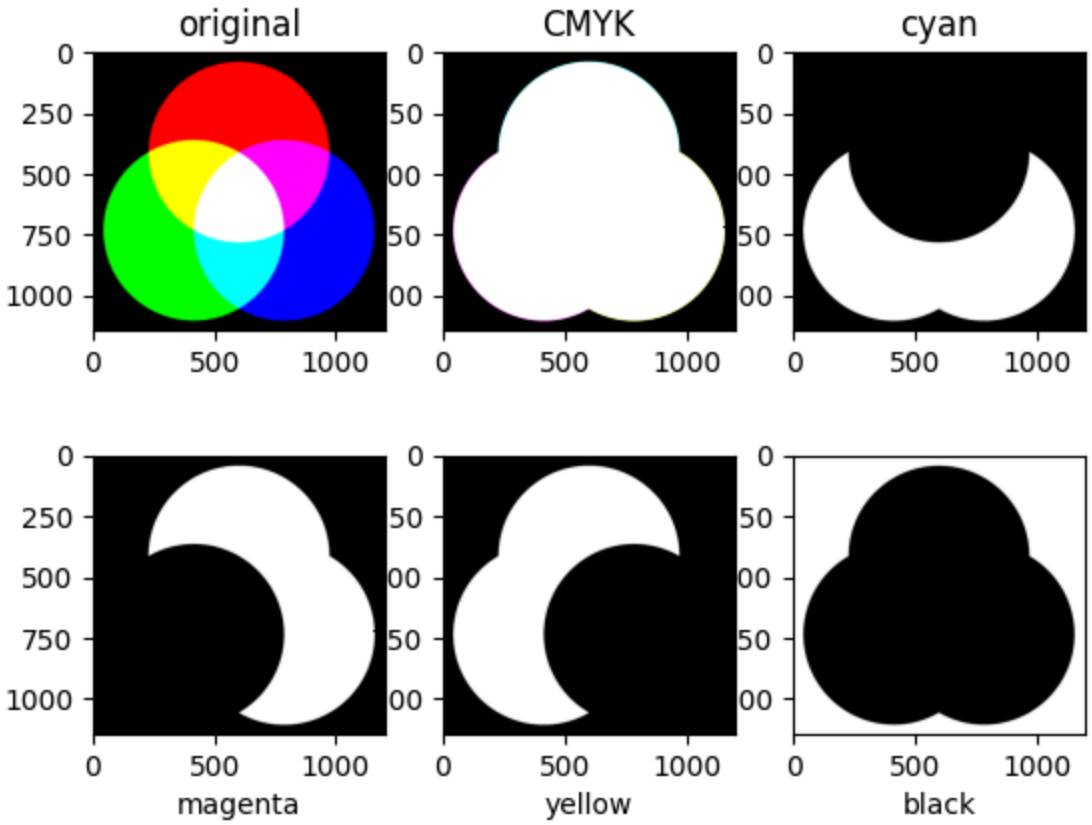


0.0.2 CONVERSIONES DEL ESPACIO DE COLOR

0.0.2 CONVERSIONES DEL ESPACIO DE COLOR

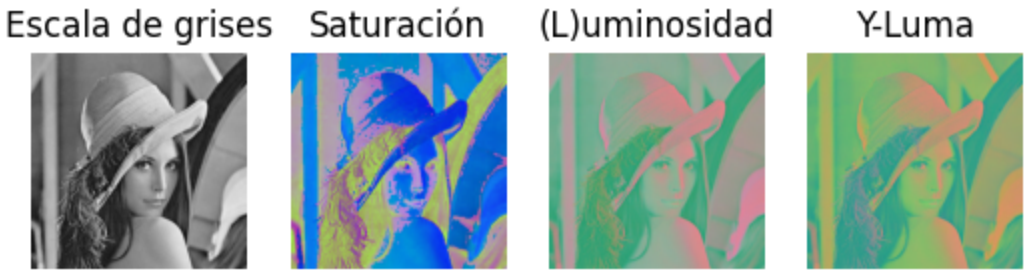
In [17]:
```
# RGB to CMYK
import numpy as np
from skimage import io

img = io.imread('images/colores.png')
rgb = img.copy() #normalizamos la imagen
rgb_p = rgb.astype('uint8')/255

with np.errstate(invalid='ignore', divide='ignore'):
    K = 1- np.max(rgb_p, axis=2) # Extrae los canales de acuerdo con la ecuación de
    C = (1- rgb_p[:,:,0]- K) / (1- K)
    M = (1- rgb_p[:,:,1]- K) / (1- K)
    Y = (1- rgb_p[:,:,2]- K) / (1- K)
```

```
    CMYK = (np.dstack((C,M,Y,K))*255).astype('uint8')
    C,M,Y,K = cv2.split(CMYK)

    fig, ax = plt.subplots(2,3)
    ax[0,0].imshow(img, cmap='gray'), ax[0,0].set_title('original')
    ax[0,1].imshow(CMYK.astype('uint8'), cmap='gray'), ax[0,1].set_title('CMYK')
    ax[0,2].imshow(C.astype('uint8'), cmap='gray'), ax[0,2].set_title('cyan')
    ax[1,0].imshow(M.astype('uint8'), cmap='gray'), ax[1,0].set_xlabel('magenta')
    ax[1,1].imshow(Y.astype('uint8'), cmap='gray'), ax[1,1].set_xlabel('yellow')
    ax[1,2].imshow(K.astype('uint8'), cmap='gray'), ax[1,2].set_xlabel('black')
    plt.show()
```



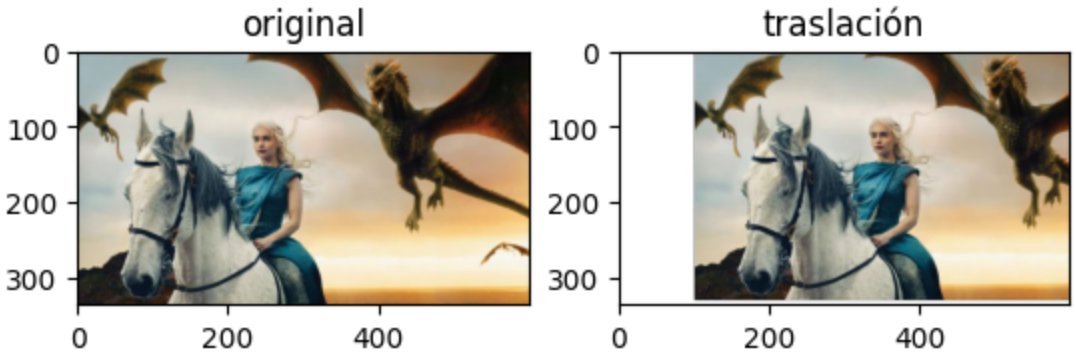
In [35]:
```
# Otras conversiones from skimage
from skimage import io
img = cv2.imread('images/Lena_RGB.png')
gray_img = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY) # gray-scale
HSV = cv2.cvtColor(img, cv2.COLOR_BGR2HSV) # (H)ue, (S)aturation and (V)alue
Lab = cv2.cvtColor(img, cv2.COLOR_BGR2Lab) # (L)uminosidad, a-b colores complementa
YCrCb = cv2.cvtColor(img, cv2.COLOR_BGR2YCrCb) # Y-Luma, Cr-Cb crominancia rojo y a

fig, ax = plt.subplots(1,4)

ax[0].imshow(gray_img, cmap='gray'), ax[0].set_title('Escala de grises'),ax[0].axis
ax[1].imshow(HSV), ax[1].set_title('Saturación'),ax[1].axis('off')
ax[2].imshow(Lab), ax[2].set_title('(L)uminosidad'),ax[2].axis('off')
ax[3].imshow(YCrCb), ax[3].set_title('Y-Luma'),ax[3].axis('off')
plt.show
```

Out[35]:
```
<function matplotlib.pyplot.show(close=None, block=None)>
```



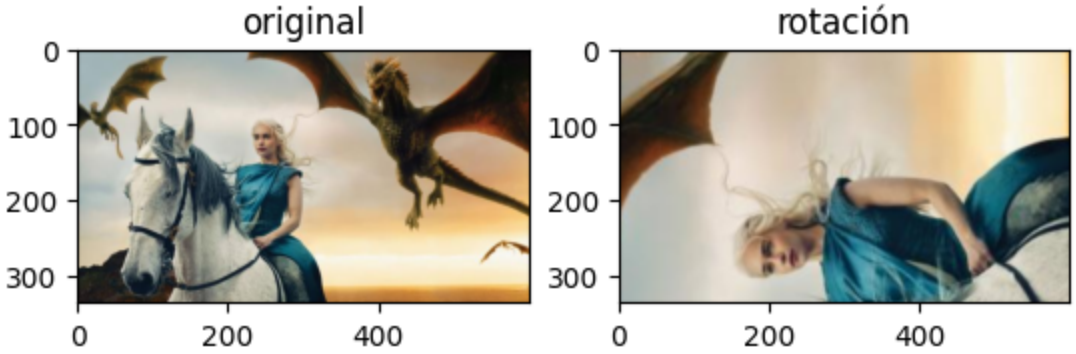
0.0.3 TRANSFORMACIONES GEOMÉTRICAS

In [41]:
```
# TRASLACIÓN
img = io.imread('images/GOT.png')
rows, cols, ch = img.shape# shape tamaño de matriz 255 rows,255 cols,3 ch

M = np.float32([[1,0,100],[0,1,-5]]) # Defino la matriz de transformación con el 30
#cuando es positvo se va a abajo y negativoo a arriba
new_img = cv2.warpAffine(img,M,(cols,rows)) # Aplico la transformación- funcion que

figs, ax = plt.subplots(1,2)
ax[0].imshow(img, cmap='gray'), ax[0].set_title('original')
ax[1].imshow(new_img, cmap='gray'), ax[1].set_title('traslación')
plt.show()
```



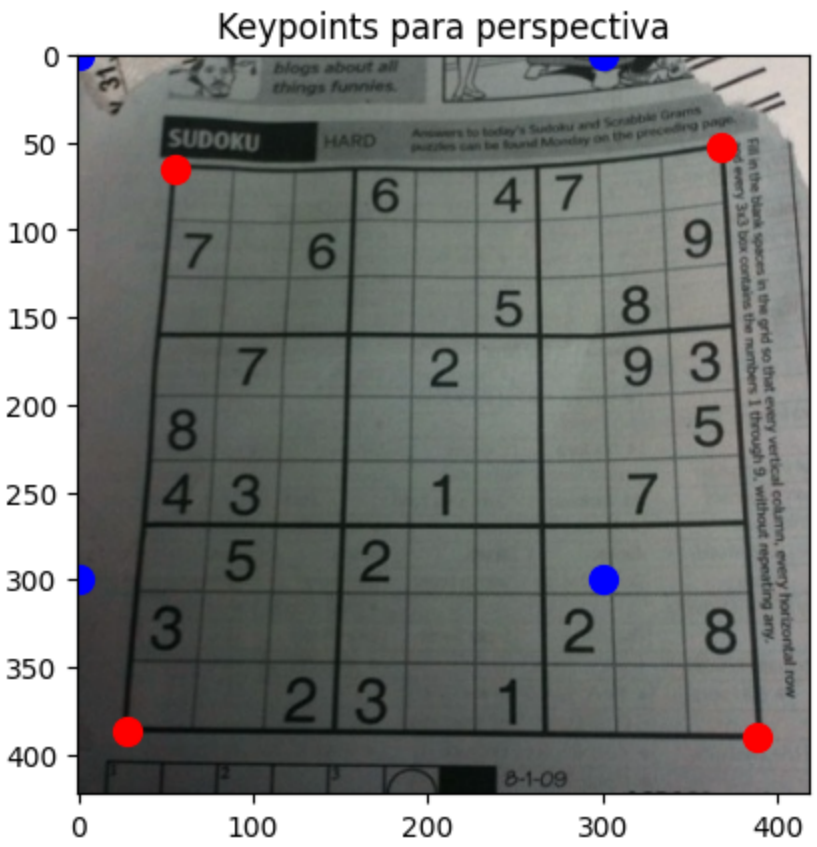
In [47]:
```
# CROPPING recorta una seccion de una imagen
img = io.imread('images/GOT.png')

new_img = img[100:290, 200:305] # en la primera le corta la x y luego la y   el img

figs, ax = plt.subplots(1,2)

ax[0].imshow(img, cmap='gray'), ax[0].set_title('original')
ax[1].imshow(new_img, cmap='gray'), ax[1].set_title('crop')
plt.show()
```

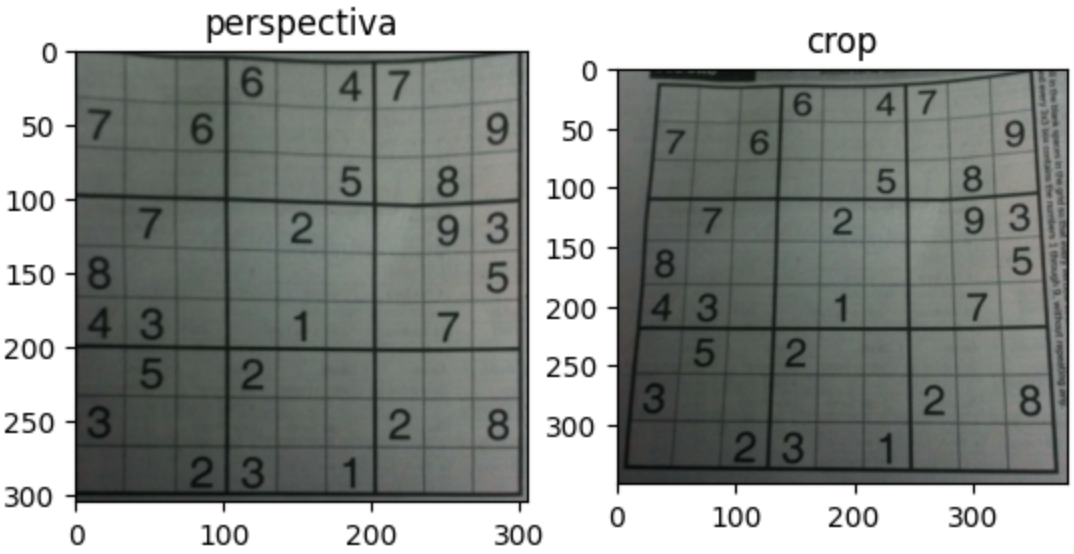
In [43]:
```
# ROTACIÓN
img = io.imread('images/GOT.png')
rows, cols, ch = img.shape

M = cv2.getRotationMatrix2D((cols/2,rows/2),angle=90,scale=2) # Defino la␣ ↪matriz
#angle=90,scale=2) esto es para qu egire la imagen
new_img = cv2.warpAffine(img,M,(cols,rows)) # Aplico la transformación

figs, ax = plt.subplots(1,2)
ax[0].imshow(img, cmap='gray'), ax[0].set_title('original')
ax[1].imshow(new_img, cmap='gray'), ax[1].set_title('rotación')
plt.show()
```



In [48]:
```
# PERSPECTIVA
img = cv2.imread('images/sudoku.png')
rows, cols, ch = img.shape
pts1 = np.float32([[56,65],[368,52],[28,387],[389,390]])
```

```
pts2 = np.float32([[0,0],[300,0],[0,300],[300,300]])

plt.imshow(img, cmap='gray')
for i in range(0,4):
    plt.plot(pts1[i,0], pts1[i,1], 'or', markersize=10)
    plt.plot(pts2[i,0], pts2[i,1], 'ob', markersize=10)
plt.title('Keypoints para perspectiva')
plt.show()

M = cv2.getPerspectiveTransform(pts1,pts2) # Defino la matriz de transformación
pers = cv2.warpPerspective(img,M,(305,305)) # Aplico la transformación
crop = img[50:400,20:400]

figs, ax = plt.subplots(1,2)
ax[0].imshow(pers, cmap='gray'), ax[0].set_title('perspectiva')
ax[1].imshow(crop, cmap='gray'), ax[1].set_title('crop')
plt.show()
```





In [49]:
```
# Leer la imagen "Lena_RGB.png" en formato RGB
img = io.imread('images/Lena_RGB.png')
# Voltear la imagen para conseguir las siguientes transformaciones. Utiliza el␣ ↪me
flipVertical = cv2.flip(img,0)
flipHorizontal = cv2.flip(img,1)
flipBoth = cv2.flip(img,-1)

figs, ax = plt.subplots(1,4)
ax[0].imshow(img, cmap='gray'), ax[0].set_title('original')
ax[1].imshow(flipVertical, cmap='gray'), ax[1].set_title('flip vertical')
ax[2].imshow(flipHorizontal, cmap='gray'), ax[2].set_title('flip horizontal')
ax[3].imshow(flipBoth, cmap='gray'), ax[3].set_title('flip ambos')
plt.show()
```